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RESEARCH ARTICLE



Examining the impact of boundary-spanning knowledge search on cooperative innovation performance among Chinese firms

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ABSTRACT

This paper examines the influence of boundary-spanning knowledge search on cooperative innovation performance and explores the moderating role of innovation passion in Chinese firms. We collected 229 valid responses from Chinese firms participating in collaborative innovation projects and applied confirmatory factor analysis to test the reliability and validity of the constructs, multiple stepwise regressions to verify the direct effect, and the PROCESS macro to test the moderating effect. The results show that all three dimensions of boundary-spanning knowledge search (i.e. network capability, IT capability, and absorptive capacity) positively affect cooperative innovation performance. Harmonious innovation passion positively moderates the relationship between network capability and cooperative innovation performance. In contrast, compulsive innovation passion plays a negative moderating role in the relationship between absorptive capacity and cooperative innovation performance. This study further enriches and extends the theoretical framework of cooperative innovation performance by emphasizing the impact of the three dimensions of boundary-spanning knowledge search.

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Boundary-spanning knowledge search; cooperative innovation performance; innovation passion; network capability; IT capability; absorptive capacity

1. Introduction

With increasing resource costs and growing uncertainty in the international market environment, it has become harder for enterprises, especially Small and Medium-sized Enterprises (SMEs), to innovate independently (Hyytinen & Pajarinen, 2007; Mao et al., 2021). Many SMEs have therefore recognized the value of collaborating with other agents, known as cooperative innovation, to overcome innovation bottlenecks (Belderbos et al., 2018; Hsieh et al., 2018; Santoro et al., 2020; Wynarczyk et al., 2013). However, agents in cooperative innovation networks exhibit different levels of cooperative innovation performance. Researchers and managers have examined this practical issue from multiple perspectives, including boundary-spanning search (Ritter & Gemunden, 2004), resource arrangement (Badrinarayanan et al., 2019), and knowledge absorption and diffusion (Zahra & George, 2002). Recent research has begun to explore the effects of knowledge management activities on cooperative innovation performance and the mechanisms underlying these effects (Attia & Eldin, 2018; Lichtenthaler & Lichtenthaler, 2009; Martinez-Conesa et al., 2017). However, studies from a boundary-spanning knowledge search perspective remain limited (Feng & Ma, 2020).

Boundary-spanning knowledge search refers to an enterprise's comprehensive ability to develop technology and create new products by acquiring heterogeneous resources through boundary-spanning examination (Zhang et al., 2011). Such stability can have negative effects. Specifically, if firms assume their boundaries should remain stable, they may lack the willingness and motivation to conduct boundary-spanning knowledge searches. As a result, they miss opportunities to discover and utilize external heterogeneous resources (Flor et al., 2018; Rosenkopf & Nerkar, 2001).

In addition, the few studies that have examined the influence of boundary-spanning knowledge search on cooperative innovation performance have generally focused on a single stage or a single dimension of boundary-spanning knowledge search (Feng et al., 2020). There is still insufficient attention to the overall mechanism that considers all dimensions of boundary-spanning knowledge search, namely, network capability, IT capability, and absorptive capacity (Zhang et al., 2011). It is therefore crucial to study these three dimensions and identify their distinct influences on cooperative innovation performance.

The concept of innovation passion is derived from entrepreneurial passion and is embedded in

a specific innovation situation (Cardon et al., 2009). This implies that enterprises' cognition of innovation can influence their innovation activities (Reisenzein, 2019). In cooperative innovation networks, agents can not only perceive the spillover effects generated by heterogeneous resources during boundary-spanning search (Gallagher, 2019) but can also experience the contagion effect of innovation passion from their partners (Story et al., 2013). By extending the dualistic model of passion (Vallerand et al., 2003) to the organizational level, we conceptualize innovation passion as a collective affective driver that spreads through inter-firm networks via emotional contagion (Cardon et al., 2017).

The stronger the innovation-related cognition and emotional factors of one agent in an innovation network, the more strongly other partners can perceive them (Ajzen, 1991; Cardon et al., 2017). In particular, one agent's harmonious or compulsive passion (Shan & Jing, 2022) toward innovation can be consciously or unconsciously transmitted to other partners during boundary-spanning knowledge interaction (Antonioli & Mazzanti, 2017). In this way, one agent's innovation passion can diffuse throughout the cooperative innovation network and, to some extent, shape other agents' boundary-spanning knowledge search behaviors (Gherardi et al., 2007; Sie & Yakhlef, 2013). Despite this, few studies have examined these dynamics in the context of cooperative innovation. Further investigation of these two forms of passion and their possible moderating roles between boundary-spanning knowledge search and cooperative innovation performance is therefore theoretically significant.

To address these gaps, this paper draws on organizational boundary theory, resource-based theory, self-determination theory, and self-regulation theory (Aldrich & Herker, 1977; Barney, 2018; Deci et al., 1994) to construct a theoretical model that links boundary-spanning knowledge search to cooperative innovation performance and incorporates the moderating role of innovation passion. The goal is twofold. First, we aim to broaden theoretical perspectives that explain differences in cooperative innovation performance among agents. Second, we aim to provide practical guidance to enterprises in innovation networks seeking to enhance cooperative innovation performance and achieve industrial transformation and upgrading.

The rest of this paper is organized as follows. Section 2 presents the theoretical background, develops the research hypotheses, and outlines the theoretical framework. Section 3 describes the research methods. Section 4 reports the data analysis and

empirical results. Section 5 concludes by discussing the main findings, contributions, and limitations.

2. Literature review and hypotheses development

2.1. Boundary-spanning knowledge search and cooperative innovation performance

Boundary-spanning knowledge search refers to the comprehensive ability of enterprises to develop technology and create new products by obtaining heterogeneous resources through boundary-crossing examination. Traditional boundary theory holds that an enterprise must maintain the stability of its boundary with the external environment (Aldrich & Herker, 1977). On the one hand, it can provide enterprises with an independent and stable environment, separated from external influences, which guarantees their autonomy to operate and develop and reduces uncertainties to some extent (Thompson, 1967). On the other hand, enterprises are inevitably exposed to the risk of knowledge leakage (Cross et al., 2000) when interacting with the external environment through information and resource flows. An enterprise's boundary can serve as a protective barrier, safeguarding its knowledge and reducing opportunistic behavior driven by market competition. However, it still has some drawbacks if the boundary is so stable. First, it leads to organizational rigidity, meaning that an enterprise relies on prior experience to address unexpected problems in its operational processes (Liao, 2002). Second, it will limit the flow of information. Although the traditional stable boundary prevents core information from spilling over, it simultaneously obstructs the entry of high-value-added information from the external environment, leaving the enterprise with limited adaptability and sensitivity to market changes (Rindova & Kotha, 2001).

This study adopts organizational boundary theory as the core analytical lens to examine heterogeneity in cooperative innovation performance. Organizational boundary theory argues that the firm's boundary is not fixed or closed but rather a permeable, agentic interface (Aldrich & Herker, 1977). Within an innovation network, a firm's ability to proactively and effectively span its organizational boundary to access critical knowledge resources is essential for gaining an advantage in cooperative activities.

Accordingly, "boundary-spanning knowledge search", the core independent variable in this study, is essentially a direct reflection of the firm's boundary management capability. It forms the foundation and point of departure for our theoretical framework. However, to conduct boundary-spanning knowledge search successfully, firms must rely on specific internal capabilities (Barney, 2018).

A boundary-spanning knowledge search can eliminate the negative impact of a stable boundary. Combining the spanner, carrier, and impetus of boundary-spanning search, three dimensions of boundary-spanning knowledge search are identified: network capability, IT capability, and absorptive capacity (Ouyang et al., 2012). Network capability refers to the ability of enterprises to identify, establish, and manage the external network (Ritter & Gemunden, 2004), IT capability refers to the ability of enterprises to allocate and utilize IT infrastructure, IT human resources, and IT intangible resources (Bharadw, 2000) and absorptive capacity refers to the ability of an enterprise to acquire, transform, and apply external knowledge (Cohen & Levinthal, 1990).

2.2. Network capability and cooperative innovation performance

First, enterprises can directly access heterogeneous resources and knowledge from the cooperative innovation network through network capability, increasing the probability of creating new products or services (Chesbrough & Crowther, 2006; Peng, 2025). Additionally, it is helpful for enterprises to cross boundaries quickly through strong network capabilities. That is, enterprises can effectively reduce the cost of searching for scarce knowledge resources and increase the probability of innovation by establishing heterogeneous knowledge resource networks and building a cooperative base among partners through strong network capabilities (Mohr & Sengupta, 2002). Second, it can reinforce the effective landing of innovation resources through network capability by coordinating the internal structure and the external relationship network (Chin et al., 2021). An enterprise with good network capability will improve its innovation performance by establishing and managing its cooperative innovation network (Ritter & Gemunden, 2004).

H1a: Network capability is positively associated with cooperative innovation performance; that is, the stronger the network capability of an enterprise, the higher its cooperative innovation performance will be.

2.3. IT capability and cooperative innovation performance

IT capability will affect cooperative innovation performance in the following ways. First, enterprises will effectively transform heterogeneous innovation resources through IT capability, improving cooperative innovation performance (Cheng et al., 2025). In other words, the enterprise can encode the knowledge

resources from internal and external innovation networks. It can effectively integrate heterogeneous knowledge resources from boundary-spanning search to obtain resource-based advantages (Sambamurthy et al., 2003).

Second, IT capability will help enterprises quickly respond to the external environment and provide technological support for cooperative innovation. According to the Dynamic Capability Theory, enterprises should continually build and reshape their innovation agility to adapt to changes in the external environment and the volatility of competitive pressure (Duan, Liu, et al., 2021). So, an enterprise is assisted in broadening its boundaries and keeping pace with agents from the innovation network by promoting and developing technology, including but not limited to infrastructure and talent training. It not only supports enterprises in responding to the environment quickly but also helps them gain a competitive advantage and benefit from cooperative innovation (Lucas & Olson, 1994).

H1b: IT capability is positively associated with cooperative innovation performance; that is, the stronger the IT capability of an enterprise, the higher its cooperative innovation performance will be.

2.4. Absorptive capacity and cooperative innovation performance

Enterprises can effectively acquire, transform, and apply external heterogeneous information resources through absorptive capacity. First, enterprises can increase their knowledge stocks and broaden their resource base by acquiring knowledge from the innovation network (Rosenkopf & Nerkar, 2001). Enterprises can not only expand the scope of knowledge search to cross-industry innovation but also avoid the waste of innovation costs with high knowledge acquisition (Cohen & Levinthal, 1990). Second, enterprises can accelerate the transfer and digestion of heterogeneous resources obtained through boundary-spanning search by transforming them. Knowledge transformation is the process by which enterprises analyze the value of innovative external knowledge through absorptive capacity (Atuahene-Gima, 2003). Afterward, it can reasonably assess the merits and demerits of heterogeneous knowledge resources to support cooperative innovation, accelerate the coupling of external and internal knowledge systems, and promote the reorganization and circulation of resources in the innovation value chain (Duan, Liu, et al., 2021). Third, it can help enterprises transform heterogeneous cooperative resources into new products or services in a timely manner by applying

knowledge. Knowledge application refers to the process enterprises use to transform innovative knowledge into practical innovation achievement through absorptive capacity (Cohen & Levinthal, 1990). New knowledge will be transformed to address current market needs if enterprises possess robust absorptive capacity.

Therefore, enterprises' cooperative innovation performance can be improved by recognizing the path of innovation development and bettering the link between the innovation value chain and the industrial chain (Zahra & George, 2002).

H1c: Absorptive capacity is positively associated with cooperative innovation performance; that is, the stronger the absorptive capacity of an enterprise, the higher its cooperative innovation performance will be.

2.5. The moderating effect of innovation passion

As an antecedent of cooperative innovation performance, innovation passion is defined as the emotion enterprises feel when engaging in innovation activities (Reisenzein, 2019). Moreover, individuals' passion influences their emotional characteristics and behavioral tendencies (Sánchez, 2019). Strong innovation passion means enterprises can recognize the importance of innovation, have a strong desire to create (Cardon et al., 2017), and actively generate creative ideas and develop new technologies to overcome the innovation dilemma.

The concept of innovation passion, while sharing conceptual roots with entrepreneurial passion (Cardon et al., 2009), is fundamentally grounded in the broader, well-established literature on work motivation, specifically in self-determination theory (Deci et al., 1994; Vallerand et al., 2003). Self-determination theory distinguishes between autonomous and controlled motivation, which provides the foundational logic for the dualistic model of passion. This theoretical extension allows us to conceptualize innovation passion not merely as an entrepreneurial trait, but as a collective affective driver that can exist within any organization engaged in innovation (Reisenzein, 2019). To explain this psychological driving mechanism, we draw on self-determination theory to further account for the role of innovation passion.

According to the Self-Determination Theory, an enterprise's willingness to innovate is also influenced by the passion for innovation of other enterprises in its vicinity (Gallagher, 2019). Specifically, if an enterprise recognizes the importance of innovation activities for its development, it will actively put them on the agenda and enter the process of autonomous internalization. Otherwise, if the enterprise pursues

innovative activities solely under external pressure, although it can still internalize them as an important decision in its development, the negative emotions generated by external stress will undermine the process of cooperative innovation performance. Harmonious innovation passion and compulsive innovation passion are defined according to the two kinds of innovation passion outlined above.

2.6. The moderating effect of harmonious innovation passion

Harmonious innovation passion refers to the cognition and emotion an enterprise is willing to innovate and internalize innovative activities (Cardon et al., 2009; Shan & Jing, 2022). Recognizing innovation's value, enterprises with a harmonious passion for innovation willingly engage in innovative activities to gain self-determination. Consequently, they become more willing to make sustained investments in innovation (Scott & Bruce, 1994). Specifically, it will be analyzed to examine the moderating role of harmonious innovation passion in the relationship between boundary-spanning knowledge search and cooperative innovation performance. First, enterprises with harmonious innovation passion are more likely to participate in cooperative relationship networks with a more open and sharing attitude (Cardon et al., 2009). When the external environment is transformed into a friendly, open resource network, enterprises with a passion for harmonious innovation may be encouraged to access a wider range of heterogeneous resources. Second, enterprises with harmonious innovation passion will be willing to learn new technologies and construct an innovation atmosphere with other agents in a cooperative innovation network (Sahoo et al., 2025). When faced with the complex process of innovation and the uncertain risks of innovation development, enterprises with positive autonomy internalization can often stimulate positive innovation passion and are more likely to promote the production of collaborative innovation outcomes (Ajzen, 1991). Third, enterprises with a passion for harmonious innovation are more active in evaluating the interplay between individual conditions and the external environment, and in coping with changes in the external environment (Farmers et al., 2003).

H2a: Harmonious innovation passion positively moderates the relationship between network capability and cooperative innovation performance.

H2b: Harmonious innovation passion positively moderates the relationship between IT capability and cooperative innovation performance.

H2c: Harmonious innovation passion positively moderates the relationship between absorptive capacity and cooperative innovation performance.

2.7. The moderating effect of compulsive innovation passion

Compulsive innovation passion refers to the cognition and emotion that an enterprise experiences when engaging in innovative activities and internalizes the process under external pressure (Cardon et al., 2009; Shan & Jing, 2022). Compared with harmonious innovation passion, the cognitive mode in which enterprises carry out innovation activities under external pressure may negatively affect the relationship between boundary-spanning knowledge search and cooperative innovation performance in the following ways.

First, enterprises with compulsive innovation passion will significantly weaken the willingness to expand network resources and actively deepen cooperation, making it more challenging to achieve good innovation performance (Idris et al., 2018). Second, compulsive passion drives short-termism. This undermines sustainable innovation mechanisms in two ways: (a) suppressing technology transformation investments, and (b) inducing psychological fatigue (Maslach & Jackson, 1981). The innovation impulse fades, especially after emotional resources are exhausted, and cooperative innovation activities inevitably lead to psychological fatigue. As a result, it is not sufficient for the initiative to build the sustainable technological environment required for cooperative innovation, which may curb the search for potential innovation opportunities. Lastly, employees with compulsive innovation passion cannot recognize the necessity of growth driven by innovation activities, will gradually lose their self-efficacy in the cooperative

innovation process, and will be unable to effectively utilize their resource endowment (Beghetto, 2006). It can not only directly inhibit employees' potential creativity but also harm their passion for innovation, leading them to lose the ability to internalize innovative knowledge and skills into achievements. As a result, it will finally influence the performance of cooperation innovation (Duan et al., 2022).

H3a: Compulsive innovation passion plays a negative moderating role in the relationship between network capability and cooperative innovation performance.

H3b: Compulsive innovation passion negatively moderates the relationship between IT capability and cooperative innovation performance.

H3c: Compulsive innovation passion plays a negative moderating role in the relationship between absorptive capacity and cooperative innovation performance.

2.8. Theoretical framework

In summary, our theoretical framework presents a coherent narrative. Organizational boundary theory underscores the fundamental importance of boundary-spanning processes. The resource-based view operationalizes this process into three distinct capability sets (network, IT, and absorptive capacities) that directly impact performance (H1a-c). Finally, self-determination theory introduces a critical contingency by proposing that the efficacy of these capabilities is either enhanced or undermined by the nature of the firm's innovation passion, harmonious or compulsive (H2a-c, H3a-c). Figure 1 visually represents this integrated model.

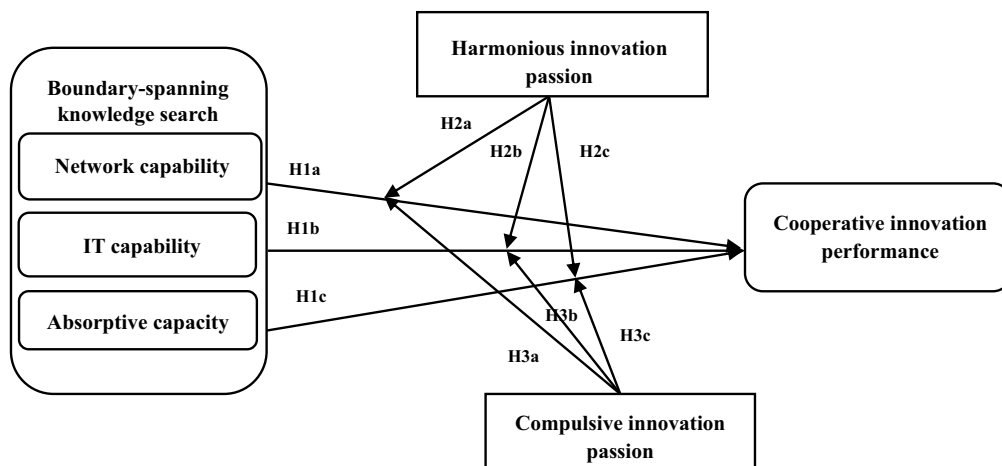


Figure 1. Theoretical framework.

3. Research methods

This study used a questionnaire to collect data, given the subjectivity of the variables in this paper and their recognition in empirical research. In the preliminary stage, this article tested the reliability and validity of the research items to ensure the scientificness and reliability of the formal research results. This study adjusted the questionnaire structure based on the constructive results obtained during this stage. In the second stage of the formal survey, we confirmed the support for the empirical testing model through reliability, validity, and correlation tests.

3.1. Research philosophy and design

With the exception of the control variables, the variables were measured on a 7-point Likert scale (1 = completely disagree, 7 = completely agree). This study used the current relatively authoritative questionnaire as a starting point, invited a professor specializing in business English and two Chinese doctoral students specializing in business English to form a relevant supervisory team, and revised the basic questionnaire in the context of the reality of the study's subject area, Yunnan Province, China, to ensure the reliability and accuracy of the study results. To improve communication, the study obtained assistance from the Yunnan Province Ministry of Science and Technology to contact the subject companies via WeChat (a common social media platform in China), phone, and email to obtain the managers' consent to participate in the study.

3.2. Research instrument and data collection

To select suitable samples, all cooperative innovation projects over the past 10 years were first searched on the official website of the Science and Technology Department of Yunnan Province, China. Given the importance of enterprises in the cooperative innovation network, this paper uses the enterprises that participated in the cooperative innovation projects above as the sample and collects data directly from their senior management and project managers. Specifically, during sample collection, to ensure the accuracy and rationality of the formal survey, we conducted a small-scale pilot study as preparatory work. The pilot survey was distributed to two main groups. On the one hand, it was distributed to students in the MBA classes of the research team's school who met the selection criteria: 1. Employed by the sample companies; 2. Holding positions of senior manager or above. A total of 30 individuals were selected, of whom 20 agreed to participate in the pilot study. On the other hand, 20 sample companies were randomly selected, and questionnaires were distributed via email. A total

of 60 questionnaires were distributed in the pilot study, and 47 were recovered, yielding an effective recovery rate of 78.3%. This process lasted for two months. We analyzed the recovered questionnaires, refined and optimized the items, and finalized the survey questionnaire. Subsequently, we conducted the formal questionnaire survey and sent it to the top managers of the sample companies by email and by offline delivery. The questionnaire content includes an introduction (briefly explaining the purpose and content of the survey to the respondents, and informing them that there are no right or wrong answers to any of the items, to ensure that respondents fill out the questionnaire truthfully), the main body of the questionnaire (items related to cross-boundary knowledge search, absorptive capacity, and the innovation performance of non-core enterprise cooperation), basic information about the respondents and their companies, and a closing statement (expressing gratitude to the respondents and inquiring about any unreasonable aspects of the questionnaire design). Finally, 50 enterprises were randomly selected as the questionnaire samples by the rolling distribution method. According to the research design by Yu and Duan (2012), six questionnaires were distributed to each enterprise, either on-site or by mail, with assistance from the Science and Technology Department of Yunnan Province. Three of these six questionnaires were completed by senior managers at each enterprise, and the remaining three were completed by managers directly responsible for the cooperative innovation project at each enterprise.

Among the 300 questionnaires distributed, 241 were returned, representing a return rate of 80.3%. After eliminating invalid questionnaires with blank or missing options, 229 valid questionnaires were returned, yielding an effective rate of 76.3%. This phase lasted for five months. Throughout the process, we strictly adhered to international and national standards and briefly introduced measures to ensure data confidentiality to participants.

3.3. Research sample

The characteristics of the sample are shown in Table 1. Specifically, 77.3% of the samples have been established for more than ten years, 62.4% of the samples were private enterprises, and 48.5% belong to the high-frequency cooperative innovation industries such as the electronic/communication equipment industry, the biomedical/chemical industry, the new energy industry, the software/IT service industry, and the electrical/mechanical manufacturing industry. 60.3% of the samples have more than 30 employees in R&D. The sample characteristics meet the requirements of this paper.

While focused on Yunnan Province, our sample covers industries aligned with China's national

Table 1. Characteristics of samples.

Characteristic	Classification	Frequency	Percentage (%)
Years	<3 years	24	10.5
	3–10 years	28	12.2
	>10 years	177	77.3
Nature	State-owned enterprises	51	22.3
	Private enterprises	143	62.4
	Others	35	15.3
Industry	The electronic/communication equipment industry	19	8.3
	The biomedical/chemical industry	34	14.8
	The new energy industry	6	2.6
	The software/IT service industry	31	13.5
	The electrical/mechanical manufacturing industry	21	9.2
	Others	118	51.5
Size	Small	73	31.9
	Medium	61	26.6
	Large	95	41.5

innovation priorities (e.g., green tech, biopharma), enhancing representativeness (National Bureau of Statistics, 2024).

3.4. Measures of constructs

To ensure the reliability and validity of the study, all scales used are from authoritative international journals and have been repeatedly verified. We invited a university professor who majored in business English to translate all the items into Chinese. Then, the translated items were reviewed by other professors specializing in business English to assess their appropriateness and adequacy and to avoid semantic ambiguity. Certain adjustments are made following their comments. We measure all the scales on a 7-point Likert scale ranging from “1” (strongly disagree) to “7” (strongly agree).

3.4.1. Dependent variable

We used the four-item scale developed by Jacob et al. (2010) to measure the cooperative innovation performance (denoted by Perf), with a sample item being “My enterprise has recently developed a growing proportion of new products through cooperation”. At the same time, we used the four-item scale developed by Sun et al. (2014) to measure the growth performance (denoted by Perfs) as the standby index for a test of the robustness of cooperative innovation performance, with a sample item being “The partner is more satisfied with the previous cooperation projects with my enterprise”.

3.4.2. Independent variables

To explore the direct impact of the three dimensions of boundary-spanning knowledge search on cooperative innovation performance, this paper treats them as independent variables. We used the four-item scale developed by Ritter (1999) to measure network capability (denoted by Net), with a sample item being “My enterprise can judge the development potential and value of different network memberships”. We used the

five-item scale developed by Ray et al. (2005) to measure IT capability (denoted by Itc), with a sample item being “My enterprise can use IT applications in management”. We used the five-item scale developed by Cohen and Levinthal (1990) to measure absorptive capacity (denoted by Abs), with a sample item being “My enterprise can use digested new knowledge for technological innovation”.

3.4.3. Moderating variables

We used the twelve-item scale developed by Vallerand et al. (2003) to measure innovation passion. Specifically, harmonious innovation passion (denoted by PH) is measured by six items, with a sample item being “My enterprise enjoys innovative activities”. Compulsive innovation passion (denoted by PC) is measured by six items, with a sample item being “It is difficult for my enterprise to control itself from utilizing innovation activities”.

3.4.4. Control variables

In this paper, years (denoted by Years), the nature of the enterprise (denoted by Nature), the industry (denoted by Industry), and the enterprise size (denoted by Size) are selected as control variables to account for differences among enterprises. It is consistent with previous research (Yu et al., 2022).

3.5. Common method bias

Linear regression was selected over alternative models (e.g., SEM) due to its superior handling of continuous moderators and its alignment with our goal of examining the incremental impact of interaction terms. We used SPSS 23.0 to test all variables involved in this study.

To eliminate common method bias, this study used Harman’s single-factor test. Firstly, we used Harman’s single-factor test to evaluate three conceptual variables in the proposed model (Podsakoff et al., 2003). The results showed that six factors with feature roots greater than 1 were obtained without rotation, and

Table 2. Multicollinearity diagnostics.

Variable	Tolerance	VIF
Net	0.372	2.685
Itc	0.538	1.860
Abs	0.614	1.628
PH	0.822	1.216
PC	0.996	1.004

*dependent variable = Perf. Net = network capability; Itc = IT capability; Abs = absorptive capacity; Perf = innovation performance; PH = harmonious innovation passion; PC = compulsive innovation passion.

the first factor explained 35.218% of the variation ($\leq 40\%$). The common method bias did not significantly affect the results of this study.

3.6. Multicollinearity diagnostics

Multicollinearity diagnostics were performed prior to regression analysis. Variance Inflation Factors (VIFs) and tolerance statistics were computed for all independent variables, with established cutoffs (VIF < 5, tolerance > 0.2) ensuring model reliability.

Diagnostic tests confirmed the absence of harmful multicollinearity: (a) VIF values ranged from 1.004 to 2.685, substantially below the critical value of 5; (b) Tolerance statistics exceeded 0.25 for all predictors.

3.7. Testing for reliability and validity

The reliability and validity tests of the constructs are given in Table 3. Generally, the construct has good internal reliability when Cronbach's alpha is greater than 0.7. The results show that Cronbach's alpha ranges from 0.889 to 0.952, significantly above the recommended level of 0.7. The KMO values for the constructs range from 0.781 to 0.901, which are considerably higher than the recommended level of 0.7.

CFA was then used to measure the constructs' composite reliability (CR) and convergent validity. As shown in Table 2, the factor loadings range from 0.699 to 0.924 (all higher than 0.6), the values of CR range from 0.860 to 0.945 (all significantly greater than 0.7), and the values of convergent validity (measured by average variance extracted [AVE]) range from 0.595 to 0.812 (all considerably greater than 0.5). The values of the square root of AVE for the constructs listed on their diagonal lines range from 0.796 to 0.890 (all greater than the maximum value of 0.754 of the correlation coefficients among the constructs in Table 3), so all the values meet their test criteria (Hair et al., 2006).

Table 3. Reliability and validity test of the constructs.

Construct	Items	Factor loading	Cronach's α	KMO	CR	AVE
Net	Net1	0.813	0.921	0.829	0.860	0.606
	Net2	0.804				
	Net3	0.775				
	Net4	0.718				
Itc	Itc1	0.782	0.889	0.844	0.880	0.595
	Itc2	0.821				
	Itc3	0.761				
	Itc4	0.788				
	Itc5	0.699				
Abs	Abs1	0.764	0.923	0.896	0.885	0.607
	Abs2	0.791				
	Abs3	0.793				
	Abs4	0.760				
	Abs5	0.786				
Perf	Perf1	0.860	0.907	0.798	0.935	0.783
	Perf2	0.896				
	Perf3	0.892				
	Perf4	0.891				
Perfs	Perf5	0.915	0.921	0.781	0.945	0.812
	Perf6	0.893				
	Perf7	0.924				
	Perf8	0.872				
PH	PH1	0.830	0.924	0.901	0.948	0.754
	PH2	0.855				
	PH3	0.866				
	PH4	0.905				
	PH5	0.845				
	PH6	0.809				
PC	PC1	0.918	0.952	0.890	0.945	0.743
	PC2	0.888				
	PC3	0.898				
	PC4	0.923				
	PC5	0.875				
	PC6	0.892				

*Net=network capability; Itc=IT capability; Abs=absorptive capacity; Perf=innovation performance; Perfs=growth performance; PH=harmonious innovation passion; PC=compulsive innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

Table 4. Correlation analyses of the variables.

Items	Years	Nature	Industry	Size	Net	Itc	Abs	PH	PC	Perf	Perfs
Years	–										
Nature	–0.265***	–									
Industry	0.036	0.255***	–								
Size	0.526***	–0.307***	–0.118	–							
Net	0.186**	–0.022	–0.025	0.230***	–						
Itc	0.178**	–0.029	–0.049	0.284***	0.659***	–					
Abs	0.165**	–0.074	–0.009	0.171**	0.550***	0.309***	–				
PH	0.060	–0.096	–0.063	0.094	0.464***	0.412***	0.412***	–			
PC	–0.200**	–0.001	–0.018	0.021	0.034	–0.002	–0.042	0.000	–		
Perf	0.142**	–0.078	–0.167**	0.276***	0.718***	0.553***	0.558***	0.348***	–0.102	–	
Perfs	0.171**	–0.069	–0.144**	0.271***	0.733***	0.559***	0.587***	0.275***	0.006	0.754**	–

*Net=network capability; Itc=IT capability; Abs=absorptive capacity; Perf=innovation performance; Perfs=growth performance; PH=harmonious innovation passion; PC=compulsive innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

3.8. Correlation analysis

According to the Pearson correlation coefficients shown in Table 4, network capability, IT capability, and absorptive capacity are all significantly and positively correlated with cooperative innovation performance, either measured by outcome performance or growth performance. Therefore, some of the above-mentioned hypotheses have been preliminarily verified.

4. Empirical analysis

This study's empirical test model comprises the main effect model, the moderating effect model, and the robustness test model. This paper used SPSS 23.0 and hierarchical regression to analyze each variable and test the main effects of three dimensions of boundary-spanning knowledge search on cooperative innovation performance, as well as the moderating effects of two dimensions of innovation passion.

As shown in the results presented in the preceding text, Cronbach's alpha ranges from 0.889 to 0.952, well above the recommended level of 0.7, indicating high internal consistency. The KMO values for the constructs range from 0.781 to 0.901, which are significantly higher than the recommended level of 0.7, confirming high sampling adequacy. The CR values range from 0.860 to 0.945, which are considerably higher than 0.7. The values of convergent validity

(measured by average variance extracted [AVE]) range from 0.595 to 0.812, all of which are significantly greater than 0.5. The square roots of the AVEs for the constructs listed along the diagonal range from 0.796 to 0.890, exceeding the maximum correlation coefficient of 0.754 among the constructs, demonstrating strong convergent validity. All indicate that effective confirmatory factor analysis can be conducted.

4.1. Main effect

When comparing the empirical results of Model 1 and Model 2 in Table 5, it can be seen that, given the control variables are constant, all three dimensions of boundary-spanning knowledge search have a positive effect on the dependent variable (β coefficients are 0.434, 0.166, and 0.265, respectively, with p -values less than 0.05). The regression model's ability to explain the dependent variable is substantially improved by adding the three independent variables (adjusted R^2 increased from 0.080 in Model 1 to 0.586 in Model 2, and the F -value rose from 5.985 to 47.195). Thus, H1a, H1b, and H1c were all verified.

4.2. Moderating effect

Using the validation approach of Cui and Wang (2017), the three dimensions of boundary-spanning knowledge search and the natural term of harmonious innovation passion were added, respectively, to Model 1. Then, Models 3, 4, and 5 were constructed in Table 6. Subsequently, the interaction terms between

Table 5. Tests of the main effect.

Control variables	Model1	Model2
Years	0.022	–0.064
Nature	0.046	0.005
Industry	–0.149**	–0.143**
Size	0.261***	0.107**
Net		0.434***
Itc		0.166**
Abs		0.265***
R-squared	0.097	0.599
Adjusted R^2	0.080	0.586
F-value	5.985***	47.195***

*Net=network capability; Itc=IT capability; Abs=absorptive capacity. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

Table 6. Test of moderating effect for harmonious innovation passion.

Variables	Model1	Model3	Model4	Model5	Model6	Model7	Model8
Years	0.022	−0.049	−0.005	−0.028	−0.048	−0.003	−0.033
Nature	0.046	−0.003	0.018	0.061	−0.010	0.019	0.054
Industry	−0.149**	−0.132**	−0.125**	−0.146**	−0.132**	−0.132**	−0.155**
Size	0.261***	0.126**	0.128**	0.198**	0.120**	0.123*	0.189**
Net		0.690***			0.690***		
Itc			0.453***			0.459***	
Abs				0.478***			0.495***
PH		0.113**	0.143**	0.130**	0.115**	0.153**	0.143**
PH*Net					0.132**		
PH*Itc						0.071	
PH*Abs							0.118**
R-squared	0.097	0.549	0.355	0.383	0.555	0.360	0.396
Adjusted R ²	0.080	0.537	0.337	0.366	0.540	0.339	0.377
F-value	5.985***	45.091***	20.341***	22.932***	39.300***	17.729***	20.672***

*Net=network capability; Itc=IT capability; Abs=absorptive capacity; PH=harmonious innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

the independent variables and harmonious innovation passion were introduced into Models 3, 4, and 5, respectively. Finally, Models 6, 7, and 8 were constructed in Table 6.

By comparing the significance of the regression coefficients of each variable in Model 3 and Model 6, it is clear that when adding the interaction term of the independent and moderating variables, the network capacity still showed a positive effect on the dependent variable (the β coefficient is constant at 0.690, with all p -values less than 0.001). The model and the regression coefficients of each variable remained significant, and the interaction term of the independent and moderating variables showed a positive effect on the dependent variable ($\beta = 0.132$, $p < 0.05$). This suggests that harmonious innovation passion plays a positive moderating role in the relationship between network capability and cooperative innovation performance, thus verifying H2a.

As shown in Model 7, when the interaction term between IT capability and harmonious innovation passion was introduced into Model 4, it did not show a significant positive effect on the dependent variable ($\beta = 0.071$, $p > 0.1$). However, IT capability and harmonious innovation passion still demonstrated positive effects on the dependent variables (β coefficients of 0.459 and 0.153, respectively, with all p -values <

0.05). This suggests that harmonious innovation passion plays no moderating role between IT capability and cooperative innovation performance, so H2b has not been verified.

When the interaction term between absorptive capacity and harmonious innovation passion is added to Model 5 in Table 6, as shown in Model 8, both natural terms continue to show significant positive effects on the dependent variables (β coefficients are 0.495 and 0.143, respectively, with all p -values < 0.05). Their interaction term has a significant positive effect on cooperative innovation performance ($\beta = 0.118$, $p < 0.05$), indicating that the greater the level of harmonious innovation passion, the stronger the positive effect of absorptive capacity on cooperative innovation performance, verifying H2c.

Similar to the steps above, Models 9–14 were constructed in Table 7 to test H3a, H3b, and H3c. By comparing the regression coefficients and the significance of Model 9 and Model 12, it was found that when the interaction term of network capability and compulsive innovation passion is added, network capability still has a significant positive effect on cooperative innovation performance (β coefficient decreases from 0.691 to 0.637, but the p -value is still less than 0.001), and compulsive innovation passion has a significant negative effect on cooperative innovation

Table 7. Test of moderating effect for compulsive innovation passion.

Variables	Model1	Model9	Model10	Model11	Model12	Model13	Model14
Years	0.022	−0.029	0.024	0.004	−0.039	0.040	0.012
Nature	0.046	−0.001	0.008	0.057	−0.020	−0.002	0.024
Industry	−0.149**	−0.134**	−0.131**	−0.154**	−0.129**	−0.110*	−0.155**
Size	0.261***	0.115**	0.103	0.180**	0.108**	0.096	0.146**
Net		0.691***			0.637***		
Itc			0.514***			0.493***	
Abs				0.534***			0.491***
PC		−0.130**	−0.125**	−0.118**	−0.134**	−0.136**	−0.139**
PC*Net					−0.177***		
PC*Itc						0.065	
PC*Abs							−0.264***
R-squared	0.097	0.553	0.348	0.382	0.581	0.418	0.447
Adjusted R ²	0.080	0.541	0.330	0.365	0.567	0.339	0.430
F-value	5.985***	45.857***	19.732***	22.842***	43.717***	22.635***	25.566***

*Net = network capability; Itc = IT capability; Abs = absorptive capacity; PC = compulsive innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

performance (β coefficient changes from -0.130 to -0.134 , and the p -value is still less than 0.05). The interaction term between them has a significant negative effect on the dependent variable ($\beta = -0.177$, $p < 0.001$), thus validating H3a.

As shown in Model 13, although the interaction term of ITC and PC had no significant effect on the dependent variable studied ($\beta = 0.065$, $p > 0.1$), both IT capability and compulsive innovation passion had a significant impact on cooperative innovation performance (β coefficients are 0.493 and -0.136 , respectively, and p -values less than 0.05). This means that compulsive innovation passion did not play a significant moderating role in the relationship between IT capability and the dependent variable, so H3b was not supported.

As shown in Table 7, the empirical results from Model 14 indicate that both absorptive capacity and compulsive innovation passion have significant effects on cooperative innovation performance (β coefficients are 0.491 and -0.139 , respectively, and p -values are less than 0.05). Their interaction term negatively affects cooperative innovation performance ($\beta = -0.264$, $p < 0.001$). This means that as compulsive innovation passion gradually weakens, the absorptive capacity of an enterprise will play an increasingly important role in enhancing its cooperative innovation performance, thus verifying H3c.

The possible reasons H2b and H3b are not verified are as follows. First, the IT capability may be unlikely to be influenced by subjective emotion. As mentioned earlier, IT capability was measured from IT infrastructure, IT human resource capability, and IT intangible resources. IT infrastructure capability is generally invested in the early stage of establishing a cooperative innovation alliance. Usually, it is not significantly increased until IT equipment is out or the enterprise's size is expanded (Dun et al., 2021), indicating that IT infrastructure capability is relatively stable and not easily subject to emotional fluctuations. So, as a subjective motivation or emotion, innovation passion may not influence IT infrastructure capability. Second, it may be hidden by the moderating effect of innovation passion on the relationship between IT capability and cooperative innovation performance. As mentioned earlier, this paper aims to explore the mechanisms by which boundary-spanning knowledge search and innovation passion influence cooperative innovation performance (i.e., focusing on the quantity of cooperative innovation), potentially at the expense of their effects on the quality of cooperative innovation, which, of course, needs to be further verified.

We theorize that this discrepancy stems from the fundamental nature of IT capability as a more structural and institutionalized resource. Unlike network capability, which is cultivated and maintained through dynamic social interactions, and absorptive capacity,

which is inherently cognitive and learning-based, IT capability is often embedded in tangible infrastructure, standardized procedures, and formalized systems (Bharadwaj, 2000). Once established, these technological systems provide a stable platform for innovation that operates with a certain degree of "functional rigidity". Their contribution to performance may be more directly tied to their technical robustness and integration, and less susceptible to the organization's fluctuating motivational states (whether harmonious or obsessive). In other words, a well-functioning IT system delivers value by its design, not by the passion of its users on any given day. This relative immunity to affective states could be what shields the IT-Performance link from significant moderation by innovation passion.

4.3. Robustness tests

To ensure the robustness of the empirical results and to test whether the selection of the dependent variable measurement method leads to estimation errors, we conduct a robustness check. Given that innovation outcomes encompass both quantity and quality, as outlined in the research design of Sun et al. (2014), this study uses the quality of cooperative innovation performance as a proxy indicator to re-estimate the empirical results presented in Tables 8 and 9.

Regarding the effects of the three dimensions of boundary-spanning knowledge search on cooperative innovation performance, Model 16 in Table 7 displays that all three dimensions of boundary-spanning knowledge search have a significant positive impact on it (β coefficients are 0.446 , 0.163 , and 0.270 , respectively, and p -values less than 0.001), so H1a, H1b, and H1c are supported.

Regarding the moderating effect of harmonious innovation passion between three dimensions of boundary-spanning knowledge search on cooperative innovation performance, Model 20, Model 21, and Model 22 in Table 7 display that harmonious innovation passion plays a positive moderating role between any dimension of boundary-spanning knowledge search on cooperative innovation performance (β coefficients for the interaction terms are 0.102 , 0.120 , and 0.114 , respectively, and all p -values are less than 0.1), so H2a, H2b and H2c are supported.

Regarding the moderating effect of compulsive innovation passion between three dimensions of boundary-spanning knowledge search on cooperative innovation performance, Model 26, Model 27, and Model 28 in Table 8 display that compulsive innovation passion plays a negative moderating role between any dimension of boundary-spanning knowledge search on cooperative innovation performance (β coefficients for the interaction terms are -0.171 , -0.255 , and -0.264 , respectively, and all p -values are

Table 8. Robustness tests of the main effect and moderating effect for harmonious innovation passion.

Variables	Model15	Model16	Model17	Model18	Model19	Model20	Model21	Model22
Years	0.009	−0.22	−0.061	−0.017	−0.039	−0.059	−0.015	−0.043
Nature	0.052	0.13	0.005	0.027	0.067	−0.003	0.027	0.061
Industry	−0.178**	−0.126***	−0.161***	−0.155**	−0.174**	−0.161***	−0.161**	−0.183***
Size	0.277***	0.082	0.145**	0.149**	0.216***	0.139**	0.144**	0.207***
Net		0.446***	0.673***			0.674***		
Itc		0.163***		0.430***			0.436***	
Abs		0.270***			0.457***			0.474***
PH			0.109**	0.156**	0.142**	0.110**	0.167**	0.155**
PH*Net						0.102**		
PH*Itc							0.120*	
PH*Abs								0.114**
R-squared	0.111	0.612	0.550	0.356	0.384	0.555	0.362	0.396
Adj-R ²	0.096	0.600	0.538	0.339	0.368	0.541	0.342	0.377
F-value	7.027***	49.845***	45.165***	20.478***	23.089***	39.434***	17.930***	20.731***

*Net=network capability; Itc=IT capability; Abs=absorptive capacity; PH=harmonious innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

Table 9. Robustness tests of the moderating effect for compulsive innovation passion.

Variables	Model15	Model23	Model24	Model25	Model26	Model27	Model28
Years	0.009	−0.036	0.017	−0.003	−0.045	0.032	0.005
Nature	0.052	0.006	0.016	0.063	−0.012	0.007	0.031
Industry	−0.178**	−0.164***	−0.161**	−0.183***	−0.159***	−0.141**	−0.184***
Size	0.277***	0.130**	0.121*	0.195***	0.124**	0.114*	0.161**
Net		0.679***			0.627***		
Itc			0.496***			0.477***	
Abs				0.519***			0.476***
PC		−0.116**	−0.120**	−0.135**	−0.104**	−0.152**	−0.155**
PC*Net					−0.171***		
PC*Itc						−0.255***	
PC*Abs							−0.264***
R-squared	0.111	0.556	0.350	0.385	0.582	0.413	0.450
Adjusted R ²	0.096	0.544	0.332	0.368	0.568	0.394	0.433
F-value	7.027***	46.350***	19.903***	23.122***	43.888***	22.212***	25.880***

*Net = network capability; Itc = IT capability; Abs = absorptive capacity; PC = compulsive innovation passion. Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted.

less than 0.001), so H3a, H3b and H3c are supported. Consequently, in addition to H2b and H3b, the statistical tests for all other models generally align with previous research.

However, it is worth noting that when the dependent variable is substituted as the quality, not the quantity of cooperative innovation performance, the moderating effect of innovation passion (either harmonious or compulsive innovation passion) on the relationship between IT capability and cooperative innovation performance changed from insignificant to significant (β coefficients of 0.120 and −0.255, respectively, p -values less than 0.001). This indicates

the hidden nature of the moderating effect of innovation passion on the relationship between IT capability and cooperative innovation performance and enterprises; the previous explanation of why H2b and H3b were not verified. Overall, the results of the hypotheses tests are shown in Table 10.

5. Discussion

This study uses organizational boundary theory, resource-based theory, self-determination theory, and self-regulation theory to examine the impact of boundary-spanning knowledge search on cooperative

Table 10. Results of the hypotheses tests.

Number	Hypothesis	Result
H1a	Network capability is positively associated with cooperative innovation performance.	Supported
H1b	IT capability is positively associated with cooperative innovation performance.	Supported
H1c	Absorptive capacity is positively associated with cooperative innovation performance.	Supported
H2a	Harmonious innovation passion plays a positive moderating role in the relationship between network capability and cooperative innovation performance.	Supported
H2b	Harmonious innovation passion plays a positive moderating role in the relationship between IT capability and cooperative innovation performance.	Not supported
H2c	Harmonious innovation passion plays a positive moderating role in the relationship between absorptive capacity and cooperative innovation performance.	Supported
H3a	Compulsive innovation passion plays a negative moderating role in the relationship between network capability and cooperative innovation performance.	Supported
H3b	Compulsive innovation passion plays a negative moderating role in the relationship between IT capability and cooperative innovation performance.	Not supported
H3c	Compulsive innovation passion plays a negative moderating role in the relationship between absorptive capacity and cooperative innovation performance.	Supported

innovation performance and to test the moderating effect of innovation passion.

Network capability, IT capability, and absorptive capacity all have a significant positive effect on cooperative innovation performance (0.695, $p < 0.001$; 0.514, $p < 0.001$; 0.532, $p < 0.001$). This indicates that stronger boundary-spanning knowledge search is associated with better cooperative innovation performance, which is consistent with prior research (Duan et al., 2023; Feng & Ma, 2020). Beyond confirming this effect, we also compare the corresponding coefficients of the three dimensions (0.434**, 0.166**, 0.265**) in Model 2. The results show that all three dimensions have a significant impact on cooperative innovation performance, and their relative importance, in descending order, are network capability, absorptive capacity, and IT capability.

H2 and H3 test the moderating effects of harmonious innovation passion and compulsive innovation passion, respectively. The results support H2a, H2c, H3a, and H3c, but do not support H2b or H3b. More specifically, harmonious innovation passion positively moderates the relationships between network capability and cooperative innovation performance (0.132, $p < 0.05$) and between absorptive capacity and cooperative innovation performance (0.118, $p < 0.05$). In contrast, compulsive innovation passion negatively moderates these same two relationships (-0.177 , $p < 0.001$; -0.264 , $p < 0.001$). IT capability does not exhibit a significant moderating effect ($\beta = 0.071$, $p > 0.1$; $\beta = 0.065$, $p > 0.1$). One possible explanation is that, in this study, the IT capability scale focuses on infrastructure capability, which is relatively stable and thus less affected by subjective emotional factors such as innovation passion (Dun et al., 2021).

In sum, the findings show that boundary-spanning knowledge search enhances cooperative innovation performance. They also show that different forms of innovation passion play distinct moderating roles in this relationship.

6. Conclusion

Although previous literature has explored the antecedents of performance differences among cooperative innovation agents, few studies have examined the impact of boundary-spanning knowledge search on cooperative innovation from an overall perspective. Apart from that, few scholars have probed deeply into the moderating role of passion on the relationship between boundary-spanning knowledge search and cooperative innovation performance. With an effort to explore the deeper reasons for the differentiated performance among cooperative innovation agents, this paper integrated organizational boundary theory, resource-based theory, self-determination theory, and

self-regulation theory and constructed the theoretical framework between boundary-spanning knowledge search and cooperative innovation performance while considering the moderating effect of innovation passion, and finally tested the proposed hypotheses based on 229 valid questionnaire samples.

All three dimensions of boundary-spanning knowledge search (network capability, IT capability, and absorptive capacity) have a significant positive effect on cooperative innovation performance. Moreover, different dimensions of innovation passion have heterogeneous moderating effects on the relationship between boundary-spanning knowledge search and cooperative innovation performance. Specifically, harmonious innovation passion can have a significant positive moderating effect on the relationship between network capability and cooperative innovation performance. However, compulsive innovation passion can have a significant negative moderating effect on the relationship between absorptive capacity and cooperative innovation performance. In contrast, both have no significant moderating effect on the relationship between IT capability and cooperative innovation performance.

6.1. Theoretical implications

This study further enriches and extends the theoretical framework of cooperative innovation performance, emphasizing the full impact of all three dimensions of boundary-spanning knowledge search. The existing literature has examined the positive impact of network capability, IT capability, and absorptive capacity on collaborative innovation separately (Duan, Liu, et al., 2021; Gilsing & Nooteboom, 2005; Ouyang et al., 2012), but it is rare for research on the whole influencing effect for all three dimensions of boundary-spanning knowledge search on cooperative innovation performance.

Also, this study effectively expands the research areas of innovation and passion. Previous researchers have mainly focused on the impact of passion on enterprise performance at the individual level. This paper introduced the definition of passion into the context of cooperative innovation by combining the findings of Vallerand et al. (2003) and Bonneville et al. (2011), and tested its moderating effect on the relationship between boundary-spanning knowledge search and cooperative innovation performance.

6.2. Managerial implications

These findings have several practical implications. First, adequate attention should be paid to enterprises to promote their boundary-spanning knowledge search capability. The empirical results showed that all three dimensions of boundary-spanning knowledge

search positively influence the quantity and quality of cooperative innovation. For this reason, enterprises should focus more on enhancing their boundary-spanning knowledge search capability. Specifically, enterprises should, firstly, actively build a resource-sharing mechanism with innovation partners to promote the quantity and quality of cooperation innovation by enhancing their network centrality. Secondly, it is essential for enterprises to continuously improve their IT resources to thoroughly probe their long-term impact on the quality of cooperative innovation. Thirdly, enterprises should constantly enhance their knowledge absorption and digestion capabilities to accelerate the internalization and implementation of external tacit knowledge.

Our findings on the critical role of harmonious innovation passion offer actionable insights for management practice. Rather than leaving passion to chance, firms can proactively cultivate it through specific human resource and leadership initiatives. According to the empirical results, harmonious innovation passion positively moderates the relationship between certain dimensions of boundary-spanning knowledge search and cooperative innovation performance. Therefore, enterprises should establish mechanisms to trigger and foster harmonious innovation passion and to eliminate or avoid the destructive effects of compulsive innovation passion in a timely manner. We propose a dual-pronged approach: one focused on fostering harmonious passion, and the other on mitigating the impact of compulsive passion.

On the one hand, distinguish and cultivate a harmonious passion for innovation. Since harmonious innovation passion positively moderates the relationships between network capability and cooperative innovation performance, as well as between absorptive capacity and cooperative innovation performance, enterprises should foster an autonomy-supportive innovation atmosphere by reducing external pressures and encouraging innovation driven by intrinsic motivation. They should reinforce the perceived value of innovation by communicating its significance for employees' long-term career development, stimulating their willingness to proactively engage in innovation. Furthermore, enterprises can establish emotion-resonance networks through regular "innovation achievement showcases" to share success stories, amplifying the contagion effect of harmonious innovation passion. For example, incorporate behavioral interview questions designed to identify candidates with intrinsic motivation for innovation. Example questions include: "Tell me about a time you worked on a creative project purely because you found it interesting or challenging", or "Describe a situation where you learned from a failed innovation attempt". This helps select individuals predisposed to harmonious passion; develop workshops that use case studies and job crafting exercises to help employees connect innovation

tasks to their personal values and strengths. This facilitates the autonomous internalization process central to harmonious passion; shift incentive structures from rewarding only successful outcomes to also recognizing valuable processes and learning. Implement policies such as Google's famous "20% time" or dedicated "innovation sandbox" periods, where employees have the autonomy to explore projects they are genuinely passionate about, without immediate pressure to deliver commercial results.

On the other hand, mitigate the negative impact of compulsive innovation passion. Given that compulsive passion undermines the promotive effect of absorptive capacity on innovation performance, even exerting a negative moderating effect, enterprises can implement specific strategies such as regularly auditing performance management systems to eliminate excessively high-pressure, zero-sum targets for innovation. Instead, set challenging yet attainable goals and emphasize team-based rewards to reduce the anxiety and social comparison that fuel compulsive passion; integrate measures of psychological well-being into routine organizational health surveys. Track metrics related to burnout, work-related stress, and psychological safety in R&D teams. A significant drop in these metrics can serve as an early warning sign of a compulsively passionate environment; create clear protocols for pausing or de-escalating projects that are causing sustained team distress. Encourage managers to mandate time off for employees showing signs of innovation burnout, signaling that sustainable performance is valued over short-term, passion-driven heroics.

6.3. Limitations and future research

First, the generalizability of the results was limited by the decision to focus on a single area, Yunnan Province, China. While this decision addressed the complexity of different sampling regions (or industries in the same region), it is impossible to claim that the empirical findings could be applied to all contexts. The negative moderating role of compulsive passion may be exacerbated in SMEs of Yunnan Province due to high power distance and resource constraints. A critical task for future research is to test this model in other Chinese provinces (e.g., highly competitive coastal regions such as Beijing and Shanghai) to determine whether and how the strength of these relationships varies with levels of economic development and institutional pressure.

Secondly, based on the available literature, this paper proposes hypotheses regarding boundary-spanning knowledge search, cooperative innovation performance, and innovation passion. Researchers are encouraged to explore other potential relationships or effects among the three variables.

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